

Research on Intelligent Traditional Chinese Medicine Prescription Model Based on Noisy-or Bayesian Network

Zhaohui Li

School of Information and Communication Engineering
Communication University of China
Beijing, China
e-mail: lizhh@cuc.edu.cn

Xiaogang Wang

Department of TCM, Beijing Hospital
National Center of Gerontology
Institute of Geriatric Medicine
Chinese Academy of Medical Sciences P.R. China
Beijing, China
e-mail: jameseward7613@hotmail.com

Wan Qiu*

School of Information and Communication Engineering
Communication University of China
Beijing, China
e-mail: qw1105@cuc.edu.cn

Dongxin Shi

School of Information and Communication Engineering
Communication University of China
Beijing, China
e-mail: sdx@cuc.edu.cn

Abstract—Traditional Chinese Medicine (TCM) is the traditional medicine of China, which played an active role in the fight against the pneumonia caused by the novel coronavirus. How to make TCM benefit the general public, and to get the prescriptions of the prestigious Chinese physician without leaving home is a problem worth studying. This is of great significance for the inheritance of TCM. In this paper, an intelligent prescribing model is designed based on the noisy-or Bayesian network. This model uses the correlation analysis method based on information entropy to obtain the network structure. The entire process greatly reduces the guidance of domain experts. The model uses the medical record data of a prestigious Chinese physician as training data, which can realize intelligent output Chinese medicine prescriptions with input symptom groups. The experimental results show a high accuracy of the model, and it can correctly simulate the diagnosis and treatment process of the prestigious Chinese physician.

Keywords- Bayesian network; information entropy; symptom; herb

I. INTRODUCTION

At present, "difficult to see a doctor" and "expensive to see a doctor" are still the thorny problems faced by Chinese health care, and the research on intelligent diagnosis and intelligent medicine prescription is expected to improve this situation.

Bayesian network [1] can express causality in an intuitive way, and has rich probabilistic expression capability and ability to deal with uncertain problems. In the medical field, people construct Bayesian network and use its inference function to perform medical diagnosis. Abroad, Heckerman [2] developed an expert system called PATHFINDER for diagnosing 63 different diseases of lymphoid tissue. Spiegelhalter [3] constructed a network called CHILD for

diagnosing infants with congenital heart disease, and the diagnostic accuracy reached the level of intermediate pediatric experts. Hope[4] developed the TakeHeartII system based on Bayesian Network to provide decision support for the diagnosis of cardiovascular diseases. In China, Bayesian network has been used for diagnosis of TCM syndromes and some results have been achieved. Sun [5] established a Bayesian network of subliminal anxiety and depression, and extracted the clinical common syndrome types of it. Yang [6] used Bayesian network for the differentiation of gastric cancer. There is no research on applying Bayesian network to clinical medicine prescription in TCM. This paper attempts to construct a Bayesian network composed of symptoms and herbs based on clinical medical record data, which can intelligently recommend prescriptions according to the symptom groups.

II. OVERVIEW OF NOISY-OR BAYESIAN NETWORK

Bayesian network is a directed acyclic graph composed of nodes (V) and directed edges (E). Nodes represent random variables, and directed edges connecting two nodes represent the causality between these two variables. When the cause node points to the result one, a conditional probability value is generated, which is a quantitative representation of the causal connection between the two nodes.

A simple noisy-or Bayesian network is shown in Fig. 1. The upper node represents the cause node and the lower node represents the result node. The directed edges in the network represent the dependency relationship between nodes, from upper nodes to lower nodes. Noisy-or Bayesian network can be used to describe the connection between n cause nodes $X_1, X_2, \dots, X_n$, and their influence Y . The influence mechanism of each cause node on the result node is independent, and the total influence is the logical OR of the

influence of each cause alone. Each cause node is sufficient to cause the result Y occurring when the other cause nodes are not occurring, if and only if all the influence mechanisms are failing at the same time, the result Y must not occur.

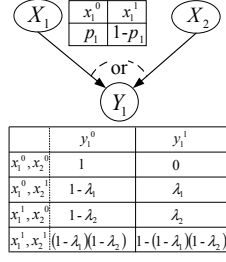


Figure 1. noisy-or Bayesian network

The premise of using the Bayesian network's inference function is to construct Bayesian network, which is to carry out structure learning [7-9] and parameter learning [10]. Structure learning is to explore the dependence of nodes in the network. Parameter learning refers to obtaining the probability distribution of each node after determining the network structure. After learning the structure and parameters of the Bayesian network, you can use the trained model to perform network inference [11]. Bayesian network inference is completed by probabilistic operations. Usually the variable of interest is set as the query variable Q , and the variable of known value is set as the evidence variable E . Reasoning is achieved by solving the probability distribution $P(Q|E)$ of the query variable.

III. METHODS

This paper designed an intelligent prescription model based on the noisy-or Bayesian network. Firstly, the clinical medical case data of the Beijing Hospital Geriatrics Department was collected, and the data was preprocessed. Then the network structure was determined combined with correlation coefficient analysis and expert experience. After that the network parameters were learned from the data. Finally the network inference was implemented on the constructed network. The entire process is shown in Fig. 2.

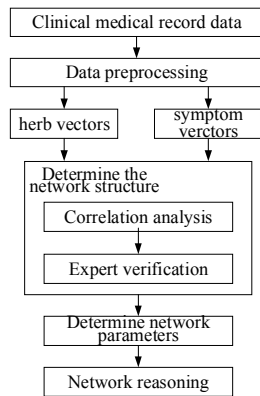


Figure 2. Block diagram of intelligent prescription Bayesian network

A. Data Preprocessing

At present, a large amount of unstructured case data has been accumulated clinically. In order to mine valuable knowledge, data preprocessing must be carried out first. The research data in this paper comes from the Geriatrics Department of Beijing Hospital and is in the form of text. The total number of cases is 1,295. A complete medical record consists the patient's treatment date, symptoms, tongue and pulse signs and prescription. The content of the medical record data is very irregular. For example, the symptom "sleeplessness" has multiple descriptions with the same meaning but different expressions: "insomnia", "excitement at night", "sleepless" and so on. According to the data characteristics, the data preprocessing scheme is designed as shown in Fig. 3.

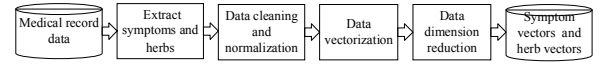


Figure 3. data preprocessing

First, the medical record document is parsed into a string, and the symptoms and herbs are extracted through string matching and string segmentation operations. Then design aggregation and split rules to standardize the names of symptoms and herbs to ensure the uniqueness of the names. Then, data conversion is performed, and it is mapped to the corresponding binary value 1, 0 according to the existence of symptoms and herbs in the medical record. Finally, dimension reduction was carried out to eliminate symptoms and herbs that were less than one-tenth of the sample size.

B. Construction of Bayesian Network for Intelligent Prescription

In order to construct the Bayesian network for intelligent prescription in this paper, we must first determine the causal structural relationship between symptoms and herbs, and then learn the probability distribution of various symptoms variables and herbs variables from the data when the structure is known.

1) Structure learning

For the structure learning of noisy-or Bayesian networks, the method of specifying the network structure by domain experts is usually adopted. For example, the PATHFINDER [3] network for diagnosing diseases of lymphoid tissues is manually constructed by developers and medical experts, which is quite time-consuming and laborious, and there are hidden dangers of strong subjectivity. In this paper, the new idea of structure learning by means of correlation analysis based on information entropy is to determine the dependence between symptoms and herbs by calculating the correlation coefficient. Assuming that there are M symptoms and N herbs in total, the specific approach is:

a) *Calculate the correlation coefficient*: For each symptom S_i , calculate the correlation coefficient between each herb H_i and it.

b) *Sort*: Sort these herbs in descending order according to the correlation coefficient value.

c) *Select*: The herb ranked in the top K has a causal relationship with the symptom S_i , and K is selected as the average number of herbs in the prescriptions.

d) *Generate directed edges*: For the symptom-herb pair with dependence, there is a directed edge in the network from the symptom to the herb.

The correlation coefficient is calculated as in (1), where $H(X_i)$ represents the self-entropy of the variable X_i , and $p(i,j)$ represents the frequency at which symptoms and Chinese medicine appear simultaneously in medical record data.

$$\mu(X_i, X_j) = \begin{cases} \frac{H(X_i) + H(X_j) - H(X_i, X_j)}{\sqrt{H(X_i)}\sqrt{H(X_j)}}, & p(i,j) > 0 \\ \frac{H(X_i) + H(X_j) - 2H(X_i, X_j)}{\sqrt{H(X_i)}\sqrt{H(X_j)}}, & p(i,j) = 0 \end{cases} \quad (1)$$

Thus, a two-layer noisy-or network composed of symptom nodes and herb nodes can be constructed, and its structure is shown in Fig. 4. The first layer of the network concludes the symptom nodes, and the second layer of the network consists of the herb ones. The directed edge in the network represents the causal relationship between the symptom node and the herb node.

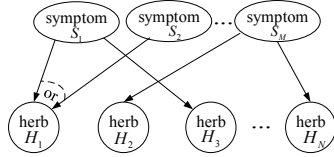


Figure 4. symptoms-herbs noisy-or network

2) Parameter learning

There are two methods for parameter learning. One is to specify network parameters by domain experts. This method is more difficult and too subjective. The other is to learn network parameters from data. The latter is used in this paper. Assuming that the nodes in the network are all binary types, the value is 0 or 1, which means that they do not occur or occur respectively. The i -th symptom node S_i of the first layer has no precursor nodes, and its probability distribution is an marginal probability distribution, which can be expressed as shown in Table I. The probability value p_i needs to be learned from the data, which can be calculated as in (2).

TABLE I. PROBABILITY DISTRIBUTION TABLE OF SYMPTOM NODE S_i

value status of symptom S_i	
$S_i=0$	$S_i=1$
p_i	$1-p_i$

$$p_i = \frac{\text{sample size without } S_i}{\text{total sample}} \quad (2)$$

For the j -th herb node H_j on the second layer, its probability distribution is the joint conditional probability distribution of all its precursor nodes, namely $P(H_j | S_1, \dots, S_i), \dots$

$q, t \in [1, M]$. Assuming that when the symptom S_q occurs, the herb H_j occurs with a certain probability λ_q , then the conditional probability distribution (CPD) of the herb node H_j can be expressed as in (3). The parameter λ_q needs to be learned from the data and calculated as in (4).

$$\begin{cases} P(H_j = 0 | S_q, \dots, S_i) = \prod_{t, S_t=1} (1 - \lambda_t) \\ P(H_j = 1 | S_q, \dots, S_i) = 1 - \prod_{t, S_t=1} (1 - \lambda_t) \end{cases} \quad (3)$$

$$\begin{aligned} \lambda_q &= P(\text{herb } H_j \text{ occurs} | \text{symptom } S_q \text{ occurs}) \\ &= \frac{P(\text{herb } H_j \text{ and symptom } S_q \text{ co-occur})}{P(\text{symptom } S_q \text{ occurs})} \\ &= \frac{\text{Sample size of occurrence of herb } H_j \text{ and symptom } S_q}{\text{Total sample}} \end{aligned} \quad (4)$$

C. Reasoning of Intelligent Prescription Bayesian Network

After learning the structure and parameters of the Bayesian network, you can use the trained model to perform network reasoning [13-15]. In this paper, using the trained Bayesian network for intelligent prescription is to solve the posterior probability problem. The design idea is as follows:

1) Set evidence variables

Set the j -th clinical symptom as the evidence variable E_j . The evidence variable has two value states: $E_j=0$ or $E_j=1$; a value of 0 indicates that the symptom does not occur, and a value of 1 indicates that the symptom occurs.

2) Set query variables

Set the i -th herb node as the query variable Q_i . Solve the posterior probability $P(Q_i=1 | E_1=e_1, \dots, E_N=e_N, e_1, \dots, e_N \in [0, 1])$, and arrange all posterior probabilities in descending order. The herb ranked in the top N is selected as one of the prescription generated by the model, and N is selected as the average number of herbs in the prescriptions.

IV. RESULTS AND ANALYSIS

A. Experimental data

After data preprocessing, 64 symptom vectors and 156 herb vectors are finally obtained, and the dimension of each vector is 1295. The symptom vector and the herb vector take the value 1 in the i -th dimension, indicating that the symptom or the herb is recorded in the i -th medical record.

B. Network structure and network parameters

By calculating the correlation coefficient of each symptom and all herbs, and selecting the herb of which the correlation coefficient value ranked in the top 14 with the current symptoms, and then submitting the result to a Chinese medicine expert for verification, a symptom-herb pair with dependence is obtained. Thus, the structure of the Bayesian network for intelligent prescription is determined. In addition, the influence strength parameter λ of the symptom-herb pair with dependence is calculated as in (4). The result is shown in Fig. 5. After the influence intensity parameters are known, the conditional probability distribution of each herb node can be obtained as in (3). In

addition, the marginal probability parameter of each symptom node are obtained as in (2), as shown in Table II.

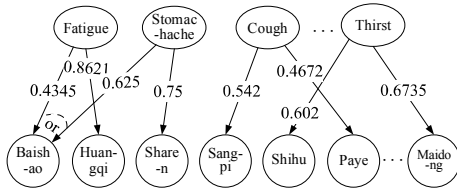


Figure 5. example diagram of network structure

TABLE II. SYMPTOM NODE'S MARGINAL PROBABILITY DISTRIBUTION PARAMETER

No.	Cause node (symptom)	Marginal probability
1	Fatigue	0.2022
2	Thirst	0.2734
3	Cough	0.1492
.....		
64	Rhinobyon	0.0432

C. Case of intelligent prescription network

After determining the structure and parameters of the Bayesian network, it is possible to infer on the constructed network to achieve intelligent prescribing. For example, we know that patient has clinical symptoms of constipation, frequent urination, and urinary incontinence, and we want to infer from the network which herbs are most suitable for composing prescription. We can separately solve the posterior probability of each herb H , as follows:

$P(\text{herb } H | \text{constipation} = 1, \text{urinary frequency} = 1, \text{urinary incontinence} = 1, \text{other symptoms} = 0)$

Then sort all herbs in descending order according to posterior probability. Finally, we select herbs ranked in the top 14 as one of the prescription generated by the model. They are Hanliancao, Fupenzi, Gouqi, Shengdi, Nvzhenzi, Sangpiaoxiao, Shanyurou, Shudi, Maren, Shanyao, Juemingzi, Zhuling, Yizhiren and Chaoqianshi.

D. Intelligent prescription accuracy assessment

The data set of this experiment is the data of 1,295 medical records in the geriatrics department. Each medical record contains a group of symptom groups and a group of herbs. The data set is randomly divided in proportion, 90% is used as the training set, and 10% is used as the test set. This experiment adopts two evaluation methods, objective evaluation and subjective evaluation to evaluate the performance of the model. Among them, objective evaluation adopts the evaluation index of accuracy rate and recall rate widely used in the recommendation system, and subjective evaluation is to evaluate the accuracy rate through expert scoring [12].

In the recommendation system, the calculation formulas of the evaluation indexes of accuracy rate and recall rate are as follows:

$$\text{precision}@n = \frac{|R(h) \cap T(h)|}{|R(h)|} \quad (5)$$

$$\text{recall}@n = \frac{|R(h) \cap T(h)|}{|T(h)|} \quad (6)$$

In the above formula, $R(h)$ represents the list of herbs for prescription generated by the model, and $T(h)$ represents the list of herbs for the original prescription in the medical records in the test set. The accuracy rate and recall rate are taken as $n=14$. The average accuracy rate and average recall rate of the model are shown in Table III. Overall, the objective evaluation results are not ideal.

TABLE III. MODEL AVERAGE ACCURACY AND AVERAGE RECALL

	Average Accuracy	Average Recall
training set	0.5618	0.5615
test set	0.5407	0.5355

The evaluation model of expert scoring is shown in Fig. 6. The score range of expert evaluation is 0~10, which indicates the degree to which the model-generated prescription fits the expert's experience.

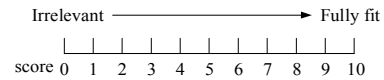


Figure 6. expert scoring model

Part of the results of expert scoring is shown in Table IV. After expert evaluation, the average score of 130 test cases was 9.82 points. The experimental results show that the noisy-or Bayesian network can rather accurately restore the diagnosis and treatment process of the prestigious Chinese physician, and can help young doctors learn the experience of the prestigious Chinese physician for diagnosis and treatment of geriatric diseases.

TABLE IV. PART OF THE RESULTS OF EXPERT SCORING

Input symptoms	Generate prescription	Score (0~10)
constipation pollakisurie incontinence	Gouqi, Shengdi, Fupenzi, Sangpiaoxiao, Hanliancao, Zhuling, Nvzhenzi, Shanyurou, Shudi, Yizhiren, Maren, Chaoqianshi, Juemingzi, Shanyao	8
fatigue edema of lower limbs shortness of breath loss of weight	Huangqi, Gouqi, Baizhu, Jiaoshenqu, Zhuling, Baishao, Shanyao, Danshen, Zexie, Houpuhua, Sangjisheng, Houpu, Dangshen, Dangguiwei	10

Although the accuracy rate and recall rate of objective evaluation are not ideal from the numerical perspective, the subjective evaluation of experts highly approves the accuracy of the model. Through analysis, it is found that there are three reasons for the inconsistency between

objective evaluation and expert subjective evaluation. First of all, clinically for a certain treatment of a certain symptom, there are a variety of similar herbs used by the prestigious Chinese physician with the same effect. The choice of herbs is replaceable. Although the generated prescription are not completely consistent with the original prescription, the actual effect is the same. Secondly, when the model generates prescription based on input symptoms, it is assumed that all the current symptoms have the same degree of priority. Clinically, each symptom may have different degree of priority, so the proportion of herb in treating each symptom may be different. Thirdly, the records of the original medical cases are not rigorous. Some of the herbs in the original prescription are superfluous, because the symptoms corresponding to them cannot be found. Therefore, the symptoms entered in the model do not include these missing symptoms, and the prescription generated by the model naturally do not contain the herbs associated with these symptoms.

As shown in Fig. 7, the input symptoms are thirst, constipation, numbness, and leg pain. The intersection of the original prescription and the generated prescription is a group of herbs that comprehensively treat the current symptoms. In addition, the Duhuo in the original prescription and the Gouji in the generated prescription both have the effect of dispelling wind-damp. They can effectively treat numbness and leg pain, and are interchangeable. Furthermore, the original prescription has two more herbs Congrong and Zhishi for treating constipation than the original prescription, indicating that in the original medical case, the symptom of constipation was an emergency. The original prescription has one less herb Shihu for thirst than the generated prescription, showing that in the original medical case, the symptom of thirst was mild. Fushen and Hehuanhua which are included in the original prescription have the effect of tranquilizing by nourishing the heart, which can treat sleeplessness, etc. However, there are no similar symptoms in the input symptoms, which may be omitted when recorded.

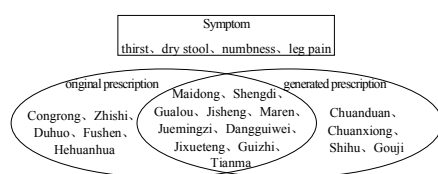


Figure 7. comparison of original prescription and generated prescription

V. SUMMARY AND EXPECTATION

In this paper, we design an intelligent prescription model based on Bayesian network. Assuming that the patient's symptoms are fully known, the noisy-or Bayesian network can better simulate the prescription process of TCM, and the prescription generated according to the symptom groups are

consistent with the characteristics of the famous TCM for the series of symptoms of geriatric disease. This model can better assist the clinical diagnosis and treatment of Chinese medicine physicians. Moreover, in the follow-up research, the weight of symptoms can also be considered, so as to prescribe herbs flexibly according to the priority of patients' current symptoms in clinical practice.

ACKNOWLEDGEMENT

This work is found by a horizontal project (NO. HG1850).

REFERENCES

- [1] N. Sophakan and C. Sathitwiriawong. Securing OpenFlow Controller of Software-Defined Networks using Bayesian Network. Proceedings of the 22nd International Computer Science and Engineering Conference (ICSEC), Chiang Mai, Thailand, 2018, pp. 1-4, doi: 10.1109/ICSEC.2018.8712754.
- [2] D Heckerman. An empirical comparison of three inference methods[J]. UAI, 1988,9(1):88.
- [3] D J Spiegelhalter, A P David, S L Lauritzen et al. Bayesian analysis in expert system s[J]. Statistical Science, 1993, 8(1):219-247.
- [4] Hope L R, Nicholson A E, Korb K B. TakeHeart II: A Tool to Support Clinical Cardiovascular Risk Assessment[R]. Tech Rep, 2007:209.
- [5] W. Sun, Y. Feng, Q. Tang. Research on TCM syndrome of Subthreshold Anxiety and Depression based on Bayesian network[J]. Chinese Journal of Traditional Chinese Medicine, 2018,33(07):3112-3115.
- [6] Y. Yang, Q. Huang, Z. Liu. Differentiation criteria for gastric cancer based on Bayesian network[J]. Journal of Shaanxi University of Chinese Medicine, 2019,42(02):119-126.
- [7] W. Manxi, W. Liandong, W. Chenfeng, G. Xiaoguang and D. Ruohai. Finding Community Structure of Bayesian Networks by Improved K-Means Algorithm[C]. in Proceedings of the 3rd International Conference on Image, Vision and Computing (ICIVC), Chongqing, 2018, pp. 865-869.
- [8] J. Suzuki. Bayesian network structure learning for discrete and continuous variables[C]. in Proceedings of the 2nd International Conference on Uncertainty Reasoning and Knowledge Engineering, Jalarta, 2012, pp. 141-144.
- [9] W. Manxi, W. Liandong, W. Chenfeng, G. Xiaoguang. Finding Community Structure of Bayesian Networks by Improved K-Means Algorithm[C]. in Proceedings of the 3rd International Conference on Image, Vision and Computing (ICIVC), Chongqing, 2018, pp. 865-869, doi: 10.1109/ICIVC.2018.8492807.
- [10] R. Jia, T. Tao and Y. Ying. Bayesian Networks Parameter Learning Based on Noise Data Smoothing in Missing Information[C]. in Proceedings of the Fifth International Symposium on Computational Intelligence and Design, Hangzhou, 2012, pp. 136-139.
- [11] L. Bennacer, Y. Amirat, A. Chibani, A. Mellouk and L. Ciavaglia. Self-Diagnosis Technique for Virtual Private Networks Combining Bayesian Networks and Case-Based Reasoning[J]. in IEEE Transactions on Automation Science and Engineering, vol. 12, no. 1, pp. 354-366, Jan. 2015, doi: 10.1109/TASE.2014.2321011.
- [12] Y. Zhang. Hidden semantic analysis technology of diagnosis and treatment data of traditional Chinese medicine[D]. East China Normal University, 2019.